

Fusion Constructive Realism (FCR v2.1)

Specification Updates & Briefing Document on Quantum Computing

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Abstract

This document formally presents the version 2.1 updates to the Fusion Constructive Realism (FCR) framework, extending the v2.0 mathematical specification. Part I details three primary architectural additions: the explicit definition of the M_n state space, the scaled phase recoil law under phase venting, and the multi-point boundary fusion postulate under metric equivalence. Part II contains an executive briefing document on Quantum Computing within FCR 2.1, specifying normalized qubit topologies, explicit mutation-driven operational gate kinetics, TikZ illustrations of CNOT evolution, and the strategic outlook for composite program executions.

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Part I

FCR 2.1 Formal Specification Updates

1 Introduction

Fusion Constructive Realism (FCR v2.0) provided a non-local, background-independent relational network framework wherein spacetime geometry and state transitions emerge from discrete correlation state spaces. FCR v2.1 introduces three critical mathematical and operational refinements to eliminate ambiguity during multi-node interactions, phase conservation, and quantum circuit operations.

2 The Three Formal Additions to FCR 2.0

2.1 Explicit Definition of the M_n Space

In FCR 2.1, we formally ground the world state Q by defining the underlying point space M_n on which metric and complex correlation structures are imposed.

$$Q = (M_n, \mu, C) \quad (1)$$

1. **Base Set (M_n):** A finite set of n discrete node identifiers:

$$M_n = \{0, 1, \dots, n-1\} \quad (2)$$

representing elementary relational points.

2. **Imposed Primitive Metric (μ):** A spatial distance mapping $\mu : M_n \times M_n \rightarrow \mathbb{R}^+$ satisfying self-distance $\mu(i, i) = 0$, rest baseline distance $\mu(i, j) = 1.0$ for non-zero correlations, and shortest-path geodesics across connected components.
3. **Imposed Complex Correlation Field (C):** An $n \times n$ Hermitian-like matrix $C(i, j) \in \mathbb{C}$ encoding complex phase and correlation strength between points $i, j \in M_n$, subject to $|C(i, j)| \leq 1.0$, $C(i, i) = 1.0$, and $C(i, j) = C(j, i)^*$.
4. **Node Capacity Axiom:** For every node $p \in M_n$, the total external capacity sum satisfies:

$$S(p) = \sum_{q \neq p} |C(p, q)|^2 \leq 1.0 \quad (3)$$

2.2 Scaled Phase Recoil Law

To guarantee net phase momentum conservation during Phase Venting ($\mathcal{M}_{1/2}$) without introducing scaling distortions across active edges of unequal magnitudes, the phase recoil absorption law is updated.

When a node $p \in M_n$ exceeds unit capacity ($S(p) > 1.0$), it sheds excess capacity $\Delta S = S(p) - 1.0$. The venting phase ϕ_{vent} is determined by:

$$\phi_{\text{vent}} = \begin{cases} \arg \left(\sum_{q \neq p} C(p, q) \right) & \text{if } \left| \sum_{q \neq p} C(p, q) \right| > \epsilon_{\text{cancel}} = 10^{-6} \\ \Phi_{\text{global}} = \arg(\text{Tr}(C^3)) & \text{otherwise} \end{cases} \quad (4)$$

Active edges $C(p, q)$ connected to p absorb a phase recoil scaled by the ratio of the total external correlation magnitude sum to the individual edge correlation strength:

$$C(p, q) \leftarrow C(p, q) \cdot \exp \left(-i \phi_{\text{vent}} \cdot \frac{\sum_{k \neq p} |C(p, k)|}{|C(p, q)|} \right) \quad (5)$$

This guarantees that smaller correlation links absorb proportionally calibrated recoil shifts, preserving phase coherence across the surrounding local network.

2.3 Multi-Point Boundary Fusion Postulate under Metric Equivalence

Open interactions between program states ($\mathcal{M}_{\text{program}}$) and data states ($\mathcal{M}_{\text{data}}$) occur via contingent fusion mutations $f \in \Sigma_{\text{contingent}}$. In FCR 2.1, boundary fusion can engage multi-node boundary subsets $\partial P_{\text{program}} \subseteq P_{\text{program}}$ and $\partial P_{\text{data}} \subseteq P_{\text{data}}$.

1. **Metric Equivalence Condition:** Multi-point boundary fusion is strictly permitted if and only if the internal pairwise metric distances within the boundary subsets match identically:

$$\forall a_i, a_j \in \partial P_{\text{program}}, \quad \mu_{\text{program}}(a_i, a_j) = \mu_{\text{data}}(b_i, b_j) \quad \text{for corresponding } b_i, b_j \in \partial P_{\text{data}} \quad (6)$$

2. **Point Identification & Weight Inheritance:** Upon fusion, boundary nodes are identified ($a_i \equiv b_i$). Overlapping or newly formed bridge correlations inherit weights according to:

$$|C_{\text{bridge}}(a_i, b_i)| = \min(|C(a_i)|, |C(b_i)|) \quad (7)$$

$$\arg(C_{\text{bridge}}(a_i, b_i)) = \phi_{a_i} \cdot \phi_{b_i} \pmod{2\pi} \quad (8)$$

Part II

Briefing Document: Quantum Computing Framework in FCR 2.1

3 Executive Overview & Foundational Qubit Topology

Quantum information processing in FCR 2.1 is modeled as relational dynamics over discrete point spaces M_n , where circuit operations correspond to discrete contingent fusion events between an operator graph ($\mathcal{M}_{\text{program}}$) and a data state graph ($\mathcal{M}_{\text{data}}$).

- **Elementary Qubit ($\mathcal{M}_2$):** The minimal qubit topology is an $\mathcal{M}_2$ link $Q_i = (p_1, p_2)$ with complex correlation $C(p_1, p_2) = re^{i\phi}$. The computational basis states $|0\rangle$ and $|1\rangle$ correspond to relative edge phase orientations $+1.0$ (0 phase) and -1.0 (π phase) relative to the global loop reference Φ_{global} .
- **Program Execution via Boundary Fusion:** Gates are physical, pre-normalized relational subgraphs in $\mathcal{M}_{\text{program}}$ that fuse with target nodes in $\mathcal{M}_{\text{data}}$ via point identification.

4 Operational Gate Kinetics

4.1 Pauli-X Gate ($\mathcal{M}_2$ Single-Qubit Operator)

- **Topology:** Single $\mathcal{M}_2$ link with fixed weight $C_X = -1.0 = e^{i\pi}$.
- **Action:** Fuses directly onto target qubit $Q = (p_1, p_2)$. Phase multiplication updates $C(p_1, p_2) \leftarrow -C(p_1, p_2)$, executing a bit flip ($|0\rangle \leftrightarrow |1\rangle$).

4.2 Phase T-Gate ($\mathcal{M}_2$ Single-Qubit Operator)

- **Topology:** Single $\mathcal{M}_2$ link carrying weight $C_T = e^{i\pi/4}$.
- **Action:** Fuses across target qubit Q , imparting a $\pi/4$ phase rotation to the active correlation edge without altering state amplitude.

4.3 Hadamard Gate (H) ($\mathcal{M}_2$ Operator on Qubit Pair)

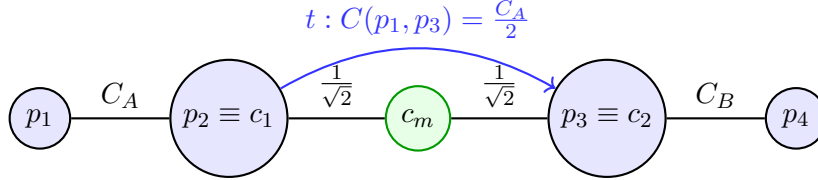
- **Topology:** Pre-normalized single $\mathcal{M}_2$ link with unit weight $C_H = 1.0$.
- **Action on Uncorrelated Nodes:** Fuses across two target nodes (p_1, p_4) . The resulting open 4-path presents two symmetric open 2-paths eligible for structural triangulation (t). Both paths carry identical transition weights $W_t = e^{-\mu}$, causing the macro state-space graph $\mathcal{M}$ to branch into two equiprobable world-states:

$$[Q_{\text{final}}] = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle \quad (9)$$

- **Action on Correlated Nodes (Execution Stagnation):** If target nodes share existing correlation paths, triangulation paths exhibit closed-cycle topologies with no open 2-paths ($W_t = 0$). This branch achieves zero transition probability, stalling execution and filtering out non-physical gate placements.

4.4 CNOT Gate ($\mathcal{M}_3$ Operator) & Mutation-Driven Phase Propagation

The CNOT operator is an open 3-node chain $P_{\text{CNOT}} = \{c_1, c_m, c_2\}$ with pre-normalized edge weights $C(c_1, c_m) = 1/\sqrt{2}$ and $C(c_m, c_2) = 1/\sqrt{2}$, ensuring $S(c_m) \leq 1.0$ is maintained throughout execution. Boundary nodes c_1 and c_2 identify directly with control node p_2 and target node p_3 via contingent fusion $f \in \Sigma_{\text{contingent}}$.



Point Identification (f) & Sequential Triangulation (t)

Pre-normalized operator edges transmit Control phase C_A directly across open 2-paths.

Figure 1: Pre-normalized CNOT operator after point-identification fusion (f), illustrating the primary triangulation step (t).

Phase transfer across the bridge occurs via a sequence of three structural triangulation mutations ($t \in \Sigma_{\text{lawful}}$):

1. **First Triangulation** ($p_1 \rightarrow p_2 \rightarrow c_m$):

$$C(p_1, c_m) \leftarrow C(p_1, p_2) \cdot C(p_2, c_m) = C_A \cdot \frac{1}{\sqrt{2}} \quad (10)$$

2. **Second Triangulation** ($p_1 \rightarrow c_m \rightarrow p_3$):

$$C(p_1, p_3) \leftarrow C(p_1, c_m) \cdot C(c_m, p_3) = \left(C_A \cdot \frac{1}{\sqrt{2}} \right) \cdot \frac{1}{\sqrt{2}} = \frac{C_A}{2} \quad (11)$$

3. **Third Triangulation** ($p_1 \rightarrow p_3 \rightarrow p_4$):

$$C_{\text{new}}(p_1, p_4) \leftarrow C(p_1, p_3) \cdot C(p_3, p_4) = \frac{C_A \cdot C_B}{2} \quad (12)$$

- If Control = $|0\rangle$ ($C_A = +1.0$), zero phase shift is transmitted; Target remains C_B ($|0\rangle$).
- If Control = $|1\rangle$ ($C_A = -1.0 = e^{i\pi}$), a π phase shift multiplies C_B by -1.0 , executing the conditional bit-flip ($|0\rangle \rightarrow |1\rangle$).

Subsequent metric expansion ($\mu \rightarrow \infty$) causes correlation decay ($d \in \Sigma_{\text{lawful}}$) to sever intermediate bridge edges below $\epsilon_{\text{sever}} = 10^{-6}$, completing the transition.

5 Strategic Outlook: Composite $\mathcal{M}_{\text{program}}$ Execution

The immediate goal for FCR 2.1 quantum information modeling is transitioning from step-by-step gate fusion to full circuit compilation within a single composite program graph $\mathcal{M}_{\text{program}}$.

Under this framework:

1. An entire quantum algorithm is compiled into a single unified operator network $\mathcal{M}_{\text{program}}$.
2. A single global fusion event merges $\mathcal{M}_{\text{program}}$ with $\mathcal{M}_{\text{data}}$.
3. A single continuous kinetic cascade of parallel triangulations (t) propagates phase values across the joint network.

4. The algorithm executes within a single macro poset transition $T \rightarrow T'$, yielding the final state class $[Q_{\text{final}}]$ directly at the observational display boundary according to the Born-rule display postulate.